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Oil-Bleeding Model for Lubricating Grease Based on Viscous Flow Through a Porous Microstructure

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One of the criteria in selecting lubricating grease for rolling-element bearing applications is its ability to bleed oil, sometimes called “grease bleeding.” Oil bleeding is assumed to be the dominating mechanism supplying new oil to the rolling track for lubrication. In this study, a physical model has been developed to understand the relation between parameters that control oil bleeding. In the model, lubricating grease is described as a porous network, formed by the thickener fibers, that contains the base oil. This type of structure is confirmed by SEM and AFM images of a lithium complex grease showing a matrix of rigid fibers with random orientation. A relatively simple flow model based on Darcy’s law for viscous flow in porous media and an anisotropic microstructure deformation model was developed. The model relates the pressure gradient, oil viscosity, thickener structure deformations, and permeability to the volumetric oil flow out of the thickener network. The permeability depends strongly on the thickener microstructure. The model was verified with experiments at a wide variety of temperatures and rotational speeds.

KEY WORDS

Oil Bleed; Microstructure Permeability; Grease Lubrication; Rolling-element Bearing

INTRODUCTION

Grease is a complex lubricant consisting of a base oil, additives, and a thickener structure that gives the grease a consistency and acts as a reservoir for the base oil. The base oil is slowly released from the grease, in a process called “oil bleeding,” and

replenishes the lubricated contact. The mechanism by which this oil release is activated is not clearly understood. In fast rotating applications, a centripetal body force acting on the grease is assumed to be the driving force. Applications like this can be found in rolling-element bearings and axial seals where an axial lip runs against a rotating flinger or vice versa. See Fig. 1.

The solid thickener in the grease is assumed to form a microstructure containing the oil. This kind of structure with a solid matrix and a fluid in the free space is often referred to as a “porous medium” (Bear (1)). In most lubricating greases used in rolling-element bearings, the thickener is a lithium hydroxystearate or lithium complex, which is also referred to as a (metallic) “soap.” Lubricating greases show a yield stress flow behavior, i.e., they show elastic behavior below a certain stress and start to flow at higher stresses. The yield stress and elasticity of the grease depend on the strength of the fibrous network and consequently on the interaction between the oil and the thickener. Fresh grease contains a soap fraction of 3–30%, but the soap fraction increases during grease life, where oil is slowly bleeding out until a critical soap fraction of 50–60% is reached at the end of grease life (Cann et al. (2)).

Baker (3) measured the oil-bleed rate of several greases at different temperatures. He found that bleeding rates increase with temperature and strongly depend on the type of base oil and thickener. It appears in his research that bearings operating at high temperatures do not require significantly higher bleeding rates to ensure long life. The type of experiments that Baker did have been normalized in the DIN 51817 (4), where the amount of oil bleed under a static load is measured after 168 h. Dynamic tests simulating oil bleed in bearing conditions have not been found in literature.

A test rig has been developed to measure oil bleeding under a centripetal body load at elevated temperatures to obtain more realistic data. The thickener microstructure is studied, and the oil flow is modeled using porous media theory. The microstructure of a lithium complex soap containing a mineral oil will be modeled and studied in more detail in the current work.

NOMENCLATURE

η	= dynamic viscosity	h	= height of grease in cup
θ	= fiber tilting angle	k	= permeability
ν	= kinematic viscosity	L	= length
ρ	= density	L_o	= initial length
ω	= angular velocity	L_{min}	= minimum length
a	= acceleration	m	= mass
A	= cross-sectional area	p	= pressure
b	= soap fiber diameter	Δp	= pressure difference
f	= fiber-volume fraction	∇p	= pressure gradient
f_o	= initial fiber-volume fraction	q	= flow velocity
f_{max}	= maximum fiber-volume fraction	Q	= volumetric flow
f_{mo}	= initial fiber-mass fraction	r	= fiber radius
F_{body}	= body force	R	= radius
$F_{friction}$	= friction force	Re	= Reynolds number
g	= gravitational acceleration	t	= time
		V	= volume
		z	= coordinate in flow direction

THEORY

Grease Microstructure

Lithium hydroxystearate soap is the most common thickener type used in rolling-bearing greases. The microstructure of this type of thickener consists of long fibers. Hurley and Cann (5) showed that a lithium hydroxystearate soap has a fine and complex network of fibers slightly twisted in some areas. Lithium complex grease has a more simple microstructure formed by longer rope-like fibers with a higher degree of twist. They used several different techniques to examine the grease microstructure and found varying values for characteristic microstructure dimensions like fiber diameter and length. We will discuss in more detail two visualization techniques: first, Scanning Electron Microscopy (SEM); and second, Atomic Force Microscopy (AFM).

Figure 2 shows an SEM image of the microstructure of a lithium hydroxystearate soap. The fibrous microstructure is clearly visible here. Making such an SEM image requires high vacuum, and the oil has to be washed out of the grease so that only thickener is left. Since grease consists of 65–95% oil, interpretations based on SEM images have to be done carefully because the final microstructure might be a result of the washing technique and the physical microstructure might have collapsed (Salomonsson et al. (6)). The advantage of AFM is that wet sam-

ples, still containing the base oil, can be examined, and it produces a true three-dimensional surface profile. See Fig. 3. However there are some limitations regarding the scanning area and field of depth, and the scanning time is much longer than for the SEM. A comparison of washed and unwashed fiber dimensions on the AFM suggests that washed fibers are slightly thicker (Hurley and Cann (5)). This is expected to be the result of the soap-solvent interaction and swelling of the soap fibers.

Soap fiber diameters found in literature using SEM and AFM range from 0.1 to 0.5 μm . Hurley and Cann (5) found a typical fiber width of 0.1 μm for a lithium hydroxystearate grease and that of 0.15 to 0.2 μm for a lithium complex grease. These values were measured from washed samples using SEM.

The fiber diameter of the lithium complex grease used in this study was measured using AFM. This is expected to give the most reliable image of the microstructure and fiber dimensions since it measures the grease directly without a washing procedure. Figure 3 shows the AFM image of the lithium complex grease. The fiber diameter is $\sim 0.2 \mu\text{m}$, and the fibers clearly have a

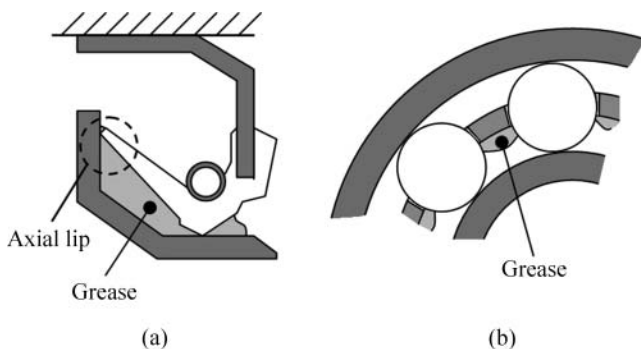


Fig. 1—Grease subjected to centripetal body load: (a) near the axial lip of a seal and (b) on the cage of a rolling-element bearing.

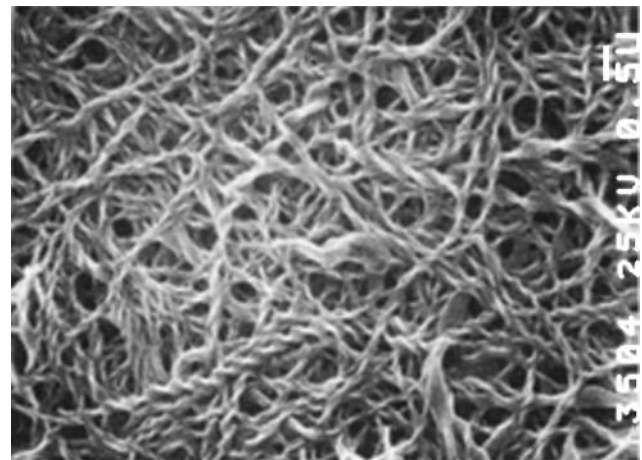


Fig. 2—SEM image of a lithium hydroxystearate soap network after washing out the base oil.

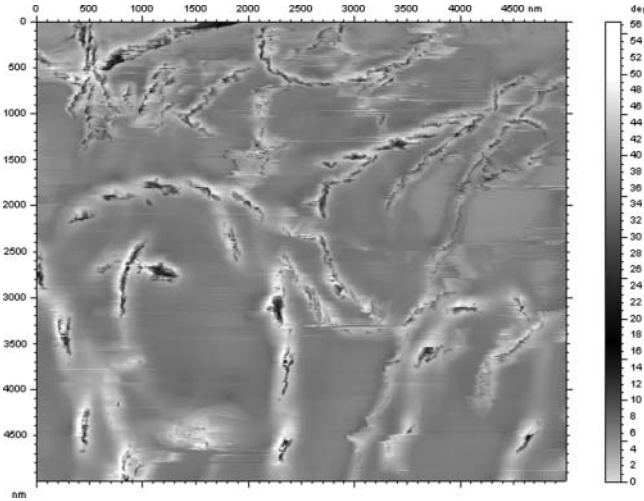


Fig. 3—AFM image of the lithium complex grease $5 \times 5 \mu\text{m}$.

twisted rope-like shape but do not show the dense and bonded structure as in the SEM image in Fig. 2.

The lithium complex grease in Fig. 3 was measured on the Atomic Force Microscope from CSM instruments using the settings in Table 1.

General Flow Equation

Fluid flow through porous media was first studied and published by Darcy in 1856 (Bear (1)). From his experiments on water flow through a fully saturated column of sand, he found a relation between the pressure difference over the column of sand and the flow rate of the water as

$$\frac{Q}{A} = \frac{k \Delta p}{\eta L}, \quad [1]$$

where Q is the volumetric flow rate through a cross-section area A . The ability of the fluid to flow through the porous media is described with the permeability constant k and the fluid viscosity η . A pressure difference Δp drives the fluid over length L in the flow direction.

Darcy's law only applies when viscous shear forces dominate the friction such that inertia effects can be neglected. This means

that the Reynolds number must be small, i.e., $Re \ll 1$ as defined by Bear (1).

For the oil-bleeding model, the oil-bleed process is assumed to be a pressure driven flow through the porous microstructure of solid soap fibers. In order to identify the different parameters involved in oil bleeding, the force balance on a small grease volume is considered:

$$\vec{F}_{body} + \vec{F}_{friction} = \Delta m \cdot \vec{a}, \quad [2]$$

where F_{body} is the total of external bulk forces acting on a volume element such as gravity and acceleration, $F_{friction}$ is the friction force on the oil due to flow through the porous soap microstructure, and the right hand side is the inertial force. To apply Darcy's law, the Reynolds number must be smaller than 1, which means that the inertial forces in Eq. [2] can be neglected. It was checked in the final model that the Reynolds number is indeed much smaller than 1 such that

$$\vec{F}_{body} + \vec{F}_{friction} = 0. \quad [3]$$

This equation is used to describe the oil bleeding of grease. The yield stress and elastic behavior of the grease are not considered here. To make a complete viscoelastic model, the geometry and deformation of the soap microstructure should be considered in too much detail to be included in the force balance. This would increase the complexity of the model enormously and is therefore not included at this time.

METHODS AND MODELING

Grease Microstructure Model

The grease microstructure has been visualized with the AFM in Fig. 3 and is shown again in Fig. 4 a. For the oil-bleed model, this measured structure is simplified to circular and randomly ordered fibers suspended in oil as shown in Fig. 4 b. For fresh grease, the fiber direction distribution is assumed to be uniform. Subsequently, to simplify the modeling of the permeability of the microstructure, the model is further simplified with the fibers ordered in an orthogonal arrangement, as shown in Fig. 4 c, with one axis parallel to the pressure gradient. This results in 1/3 of the fibers being parallel and 2/3 of the fibers being perpendicular to the oil flow. The orthogonal fiber arrangement has the advantage that anisotropy, i.e., fiber orientation, is relatively easy to model.

Base oil will flow out of the grease during oil bleed, which results in an increase in soap volume fraction. This means that the soap structure has to become denser and that the permeability will decrease, i.e., the resistance of the oil to flow through the soap microstructure increases. This mechanism makes the oil bleed decrease in time.

For the model, a homogeneous soap volume fraction over the whole grease volume is assumed. This is approximately true for sufficiently small grease volumes. Grease samples with bigger volumes that have been subjected to centripetal loads show an increase in soap fraction at the smaller radius. This can be explained with the oil flow in radial direction. For modeling the oil bleeding, the homogeneous soap distribution is assumed, and it will appear in the simulation results that this assumption is justified.

TABLE 1—SETTINGS FOR THE AFM

AFM CSM Instruments		
Operation mode	Noncontact	
Tip type	ACLA	
- Tip length	225	μm
- Oscillating frequency	160	kHz
- Free amplitude	150	nm
- Set point	30	%
Controller settings		
- P	0.06	
- I	200	μs^{-1}
Scanning speed	0.1	line/s
Scanning resolution	514×514	pixels

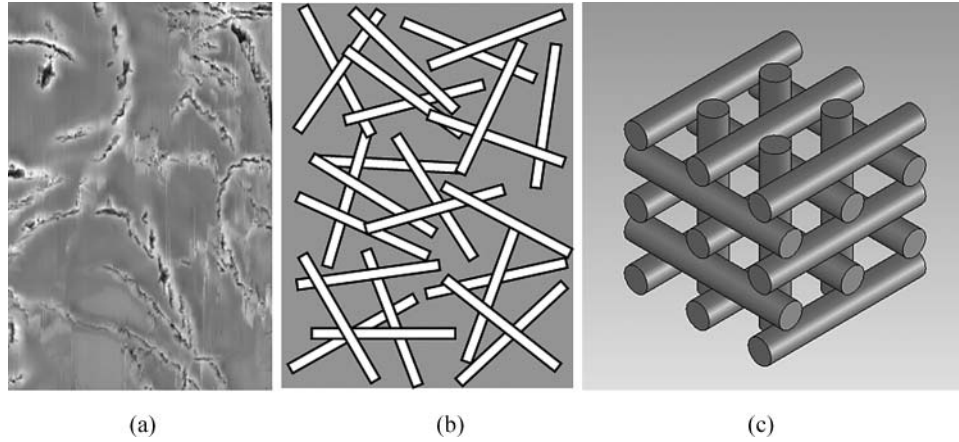


Fig. 4—Thickener structure: (a) picture made with AFM, (b) 1st simplification to rigid fibers, and (c) 2nd simplification to orthogonally arranged rigid fibers where the structure consists of stacked cubic unit cells of volume $V_o = L_o^3$ at $t = 0$.

Friction Force

The friction term in Eq. [3] is described using Darcy's law from Eq. [1], which reads in its general form as

$$\vec{q} = \frac{1}{\eta} \vec{k} \cdot \vec{\nabla} p, \quad [4]$$

where $\vec{q}$ is the fluid velocity vector, $\vec{k}$ the permeability tensor, and $\vec{\nabla} p$ the pressure gradient. From this equation, the friction force per unit volume is deduced as

$$\vec{\nabla} p_{friction} = \eta \vec{k}^{-1} \cdot \vec{q}. \quad [5]$$

This is the basic equation used in the force balance in Eq. [3] to calculate the oil flow velocity q . Eq. [5] requires values for the base oil viscosity η and the permeability k .

The base oil viscosity follows Newtonian behavior for the low oil flow velocities in the grease and is assumed to depend on temperature only. Walther's equation is used to describe the temperature–viscosity behavior (Sánchez-Rubio et al. (7)):

$$\log[\log(v + 0.7)] = A - B \log(T), \quad [6]$$

where v is the kinematic viscosity in cSt, and T is the temperature in Kelvin. A and B are constants that can be determined from two measured viscosities at known temperatures, normally 40°C and 100°C. Eq. [6] can be written for the dynamic viscosity as

$$\eta = \rho 10^{-6} \left[10^{(10[A - B \log(T + 273)])} - 0.7 \right] \quad [7]$$

where η is the dynamic viscosity in Pa·s at temperature T in °C, and ρ is the oil density.

The permeability strongly depends on the porosity of the soap microstructure. In some cases, the permeability of a porous medium can be determined directly from experiments. However, in most cases, permeability models are required to calculate the flow as a function of the pressure gradient. The same applies to oil bleeding of grease. When a good model for the grease permeability is available, the oil bleeding can be predicted.

Gebart (8) derived analytical expressions for the permeability of flow that is parallel or perpendicular to a grid of

uniaxial aligned fibers. In the orthogonal arrangement of fibers as assumed in Fig. 4 c, the combined permeability is taken as a summation of these contributions as

$$k = \frac{2}{3} k_{\perp} + \frac{1}{3} k_{//}, \quad [8]$$

where $k_{\perp}$ is the permeability perpendicular, and $k_{//}$ is the permeability parallel to a grid of uniaxial aligned fibers. The expressions for permeabilities that are parallel and perpendicular as derived by Gebart (8) are respectively

$$k_{\perp} = \frac{16}{9\pi\sqrt{2}} \left(\sqrt{\frac{f_{\max}}{f}} - 1 \right)^{5/2} r^2 \quad [9]$$

$$k_{//} = \frac{8}{57} \frac{(1-f)^3}{f^2} r^2, \quad [10]$$

where $f_{\max}$ is the maximum fiber-volume fraction, and r is the radius of the soap fibers. Eq. [9] has originally been derived for relatively high volume fractions where the pressure gradient over the complete volume is dominated by the pressure drop over the small gap region between two fibers. Gebart (9) showed that the error between his analytical equation, Eq. [9], and numerical simulations is smaller than 10% for fiber-volume fractions $f > 0.35$. For fresh grease with an initial soap volume fraction of, e.g., 0.10 and 0.20, the error is 30% and 20% respectively. The model simulation results will show whether this error at low fiber-volume fractions is acceptable for modeling oil bleeding of grease.

From Eq. [9] and Eq. [10], it is clear that as the volume fraction increases during oil bleed, the permeability decreases, which is the main reason why the oil-bleed rate tends to decrease over time. Change of the grease microstructure might cause additional reduction of the oil-bleed rate. The change of the grease microstructure can be modeled in several ways.

First the effect of fiber-volume fraction increase is studied assuming that the orthogonal arrangement is maintained during oil bleed. In this case the fiber-volume fraction increase can be achieved by increasing the fiber radius r or by decreasing the size L of the cubic unit cell. Only decrease of the cubic cell is taken into account since gradual growth of the fiber radius due to oil bleed is most unlikely to happen in reality. This means that the

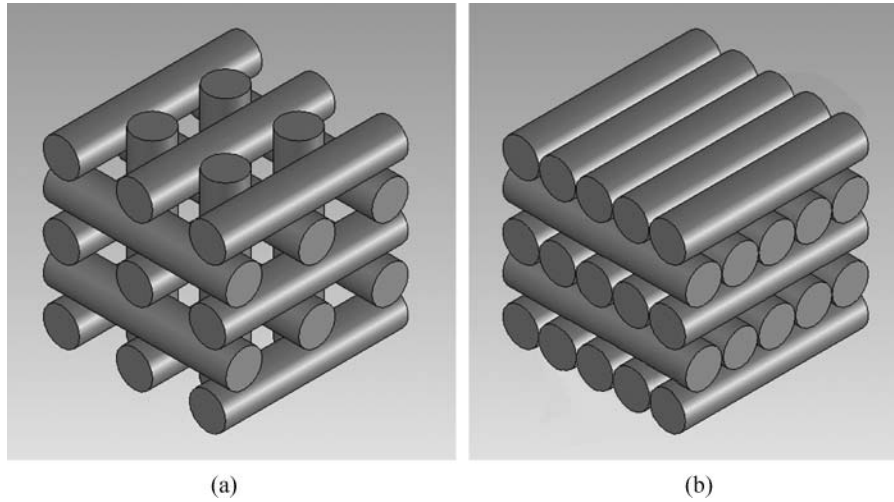


Fig. 5—Two fiber arrangements for their maximum fiber-volume fraction: (a) the orthogonal arrangement and (b) the arrangement where the fibers that were initially parallel have tilted so as to be perpendicular.

fiber radius r in Eq. [9] and Eq. [10] remains constant while f increases in time. The maximum fiber-volume fraction possible for the orthogonal arrangement is reached when the fibers are touching (see Fig. 5 a), and $f_{max} = 3\pi/16$.

Secondly, a model is proposed that assumes that fibers initially parallel to the pressure gradient gradually tilt and finally become perpendicular to the pressure gradient and oil flow. Consequently, the microstructure becomes anisotropic during oil bleed, and a higher maximum volume fraction can be reached. Figure 5 b shows the situation where all fibers have tilted so as to be perpendicular at the maximum fiber volume fraction $f_{max} = \pi/4$, which is much higher than the case of the isotropic fiber arrangement in Fig. 5 a.

The anisotropic fiber arrangement induced that Eq. [8] has to be modified to include the tilting of fibers. Figure 6 shows one

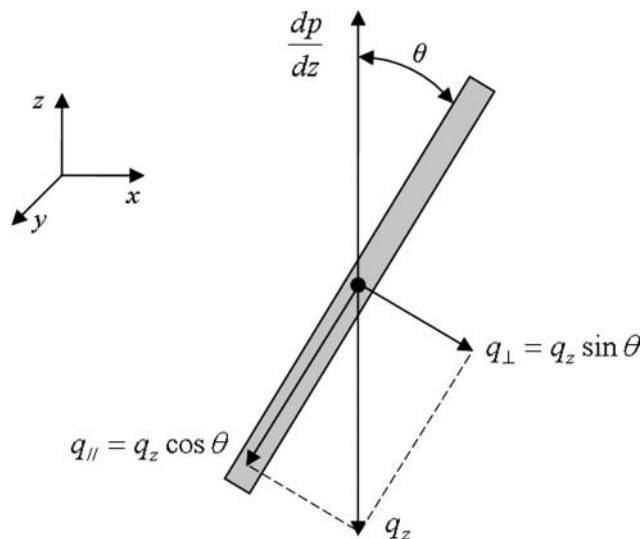


Fig. 6—Fiber tilted with angle θ and flow q components perpendicular and parallel to the fiber.

individual fiber of the group of uniaxial aligned fibers that were initially parallel to the flow but that have now tilted to an angle θ . The oil flow around this group of fibers can be decomposed into two contributions: one for flow parallel to the fibers $q_{//}$ and one for flow perpendicular to the fibers $q_{\perp}$. The pressure drop dp/dz can subsequently be written as the summation of the parallel and perpendicular flow contributions:

$$\frac{dp}{dz} = \frac{\eta}{k_{\perp}} q_{\perp}^2 \sin^2 \theta + \frac{\eta}{k_{//}} q_{//}^2 \cos^2 \theta. \quad [12]$$

This can be written as Darcy's equation:

$$q_z = \frac{k(\theta)}{\eta} \frac{dp}{dz}, \quad [13]$$

with the permeability of the tilted fibers written as

$$k(\theta) = \frac{k_{//} k_{\perp}}{k_{//} \sin^2 \theta + k_{\perp} \cos^2 \theta}. \quad [14]$$

Assuming that in fresh grease, 1/3 of the fibers is orientated parallel, and 2/3 are orientated perpendicular to the oil flow, as in Fig. 4 c, one can calculate the total permeability including anisotropy as

$$k = \frac{2}{3} k_{\perp} + \frac{1}{3} k(\theta). \quad [15]$$

To evaluate the angle θ in Eq. [14], it is assumed that the volume reduction of the cubic unit cell comes only from a reduction in height $L(t)$ such that the base surface is constant and equals L_o^2 . Consequently $L_o - L_{min}$ is the height over which the fibers tilt during oil bleed, defined as the difference between the initial volume height L_o and the minimum volume height L_{min} of the cubic cell when the maximum volume fraction is reached. The height of the fibers parallel to the oil flow at time t equals $L(t) - L_{min}$, such that the angle θ can be written as

$$\cos \theta(t) = \frac{L(t) - L_{min}}{L_o - L_{min}}. \quad [16]$$

By expressing the unit cell volume $V(t)$ as $L_o^2 L(t)$ and using the relation $V(t) f(t) = V_o f_o$, Eq. [16] can be written as a function of

the soap volume fraction as

$$\cos \theta(t) = \frac{f^{-1} - f_{\max}^{-1}}{f_o^{-1} - f_{\max}^{-1}}. \quad [17]$$

Initially, when $f = f_o$, θ equals 0° and $k(\theta)$ becomes $k_{//}$, meaning that one third of the fibers is parallel to the oil flow and that Eq. [15] becomes equivalent to the isotropic case as in Eq. [8]. During oil bleed, the angle θ will increase and will finally become 90° when f equals the maximum fiber-volume fraction $f_{\max}$. Now all fibers have tilted perpendicular to the pressure gradient, and the oil bleed stops.

Body Force

The body force consists of gravity force and centripetal force due to the rotational speed. This force builds up the pressure in the grease that pushes out the oil. The body forces acting on grease rotating with angular velocity ω at distance R around a central point are the gravity force and the centripetal force due to rotation:

$$\vec{F}_{body} = m\vec{g} + m\omega^2\vec{R}. \quad [18]$$

For the grease in a rotating application like the axial seal lip, the gravity force is smaller by at least a factor of 100 than the centripetal forces and has therefore been omitted in the model.

The pressure gradient as derived by considering the body forces on an infinitesimally small volume element located at distance R from the center of rotation becomes

$$\nabla p_{body} = \rho\omega^2 R, \quad [19]$$

where ρ is the oil density. This expression is used as the body force that drives the oil flow in Eq. [3].

EXPERIMENTS AND VALIDATION

The standard test for measuring oil separation is described in DIN51817 (4). Here a dead weight is put on top of a cup filled with grease, and the oil is slowly pressed out through a sieve at the bottom side. The mass of the oil separated from the grease is measured after resting for 1 week in an oven at 40°C . A modified test is sometimes performed at elevated temperature or higher internal pressure.

The conditions in the DIN51817 test are not representative for grease experiencing high centripetal forces resulting from high rotational speeds. A more representative test rig was designed to measure oil separation from grease under centrifugal motion at different temperatures and rotating speeds. A schematic representation of this rig is shown in Fig. 7 and the main dimensions are given in Table 2.

The cylindrical cup is filled with grease, and four of these cups are mounted on a central rotating shaft. The oil bleeding out of the cup is collected in a small retrieval container that is screwed onto the cup. This container is not shown in Fig. 7. The mass of the oil in these containers determines the oil bleed, which is measured during a test period of 24 h at a defined test temperature and rotational speed.

In the test rig, the oil flow is parallel to the centrifugal body load resulting in a relatively simple one-dimensional flow. This one-dimensional flow is modeled with the force balance in Eq. [3].

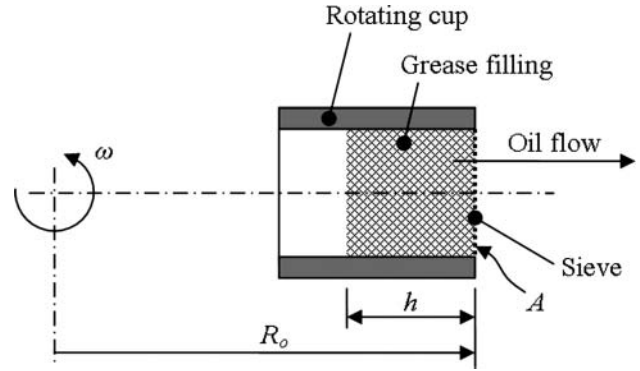


Fig. 7—Sketch of the test rig to measure oil separation from grease under centrifugal motion.

The grease selected for this study is a lithium complex grease and has a NLGI 3 consistency. This grease has a simpler microstructure than a lithium hydroxystearate grease (Hurley and Cann (5)) and is expected to be described better by the microstructure model. The fresh grease has a high percentage of thickener content, which minimizes the error for low fiber-volume fractions in the analytical permeability model, Eq. [9].

Algorithm to Calculate Oil Bleeding

A numerical calculation is used to solve the equations and predict the oil bleed in time. In each time step Δt , the oil-bleed rate is calculated as follows:

1. The fiber-volume fraction f is calculated as $f = f_0 V(t)/V_0$.
2. The permeabilities $k_{//}$ and $k_{\perp}$ are calculated using Eq. [9] and Eq. [10].
3. The angle θ and the permeability $k(\theta)$ are calculated from Eq. [17] and Eq. [14] respectively ($\theta = 0$ in case of isotropic bleeding).
4. The grease permeability k is calculated from Eq. [15] as the combination of permeabilities parallel and perpendicular to the soap fibers.
5. The pressure gradient over the grease length h in the rotating cup (Fig. 7) is calculated from Eq. [19] as either

$$\nabla p_{body} = \frac{\omega^2 \rho}{h} \int_{R_o-h}^{R_o} R dR \quad [20]$$

or, when solving the integral,

$$\nabla p_{body} = \frac{\omega^2 \rho}{2h} [R_o^2 - (R_o - h)^2]. \quad [21]$$

6. Now the oil-bleed rate can be found from Eq. [1] as

$$\frac{dV}{dt} = -Q = -A \frac{k}{\eta} \nabla p_{body}. \quad [22]$$

7. The grease volume at time $t + \Delta t$ becomes

$$V(t + \Delta t) = V(t) + \frac{dV}{dt} \Delta t. \quad [23]$$

8. The total mass percentage of oil loss is calculated from the bleeding rate and will be compared with the experimental results.

TABLE 2—TEST RIG DIMENSIONS

Test rig dimensions			
R_o	67e-3	m	outer radius
A	3.14e-4	m ²	outflow area
h_{cup}	31e-3	m	cup length

The input for the model comes from the grease properties. The grease density and base oil viscosities are given by the grease manufacturer. The fiber diameter is determined from the AFM image. The fiber mass fraction and base oil density can be measured using standard measuring methods where the oil is separated from the soap. Values for the lithium complex grease are shown in Table 3.

For the experimental tests, four different ambient temperatures and two different rotational speeds are used.

RESULTS AND DISCUSSION

The oil-bleed model contains several equations for the permeability and the grease microstructure. The flow of the base oil through this microstructure is very sensitive to the permeability, which decreases with increasing fiber-volume fraction.

Figure 8 shows the different models proposed for the permeability. The boundaries of the models are given by the two cases where all fibers are uniaxially aligned and either perpendicular or parallel to the pressure gradient, i.e., with permeability equal to $k_{\perp}$ or $k_{\parallel}$ respectively.

It can be seen here that for the perpendicular arrangement, the permeability is smallest and goes to zero at $f = \pi/4$, which was defined as the maximum fiber-volume fraction. For the parallel case, no such limit appears in the equation. This means that even at the maximum fiber-volume fraction, a flow of oil is still possible.

Consider the first case, as described in Eq. [8], where the fibers stay in the orthogonal arrangement. Figure 8 shows that the permeability is lower than for the uniaxial parallel fiber arrangement but follows the same trend.

Interesting is the second case, where the fibers that are initially parallel, tilt during oil bleed and finally become perpendicular according to Eq. [15]. Then the permeability decreases to values similar to the uniaxial perpendicular case. The permeability finally goes to zero, which means that the oil bleeding stops while the microstructure still contains oil.

The permeability derived for fibers perpendicular to the oil flow, Eq. [9], has to be used with care because this equation was originally derived for fiber-volume fractions $f > 0.35$. The grease used in the current study has an initial fiber-volume fraction of

TABLE 3—GREASE PROPERTIES

Grease properties			
ρ_{grease}	930	kg·m ⁻³	grease density
ρ_{oil}	870	kg·m ⁻³	base oil density
η_{40}	98.5	Pa·s	base oil viscosity at 40°C
η_{100}	10.5	Pa·s	base oil viscosity at 100°C
b	200	nm	fiber diameter
f_{mo}	0.26	-	fiber mass fraction

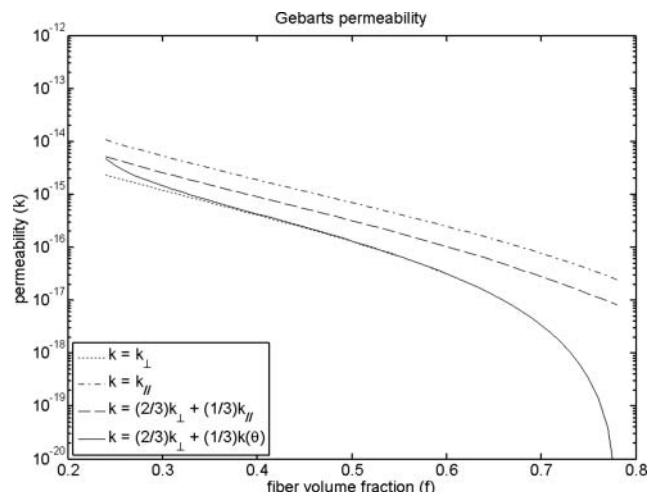


Fig. 8—Permeability models for fluid flow through the fibrous soap microstructure.

0.24, but this can be lower for other greases. However, in the initial orthogonal structure also, parallel fibers are present such that the model is not only depending on Eq. [9]. In addition, during oil bleed, higher fiber-volume fractions are reached where Eq. [9] can be used without problems.

The simplest model is considered where the grease microstructure is assumed to be isotropic such that Eq. [8] applies and the oil bleed will decrease in time due to the decrease in permeability. Here this decrease only comes from the increasing fiber-volume fraction. Figure 9 a and Fig. 9 b show the percentages of oil loss respectively at different temperatures and at different speeds. The symbols represent the experiments, and the lines represent the numerical model. The model shows trends similar to those of the experiment but predicts higher values for the oil loss and oil-bleeding rate. The isotropic model, using Eq. [8] for the total permeability, overestimates the experimental results. This means that the predicted permeability is too high and has to decrease faster with oil loss. Finally, when the maximum fiber-volume fraction is reached, the oil bleed has to stop. This, however, does not happen here since the permeability does not become zero at high fiber-volume fractions as was shown in Fig. 8.

The stopping of oil bleeding is fulfilled in the anisotropic model where the fibers that are initially parallel with the pressure gradient slowly tilt perpendicular during oil bleed. Here the oil-bleed rate will reduce faster, and there is a limit for the maximum percentage of oil loss when the maximum fiber-volume fraction is reached.

Figure 10 a shows that the results obtained with the anisotropic model using Eq. [15] predicts the oil loss with better accuracy. Here the oil loss at 120°C is still overestimated in the beginning but finally approaches the measured values. At lower temperatures, 40°C and 60°C, the first hours are described very well but the model somewhat overestimates the oil loss at later times. Also at lower speeds of 1000 rpm, as shown in Fig. 10 b, the anisotropic model describes experimental results reasonably well. Overall, the model seems to capture the main effect of speed and temperature on the oil-bleeding rate.

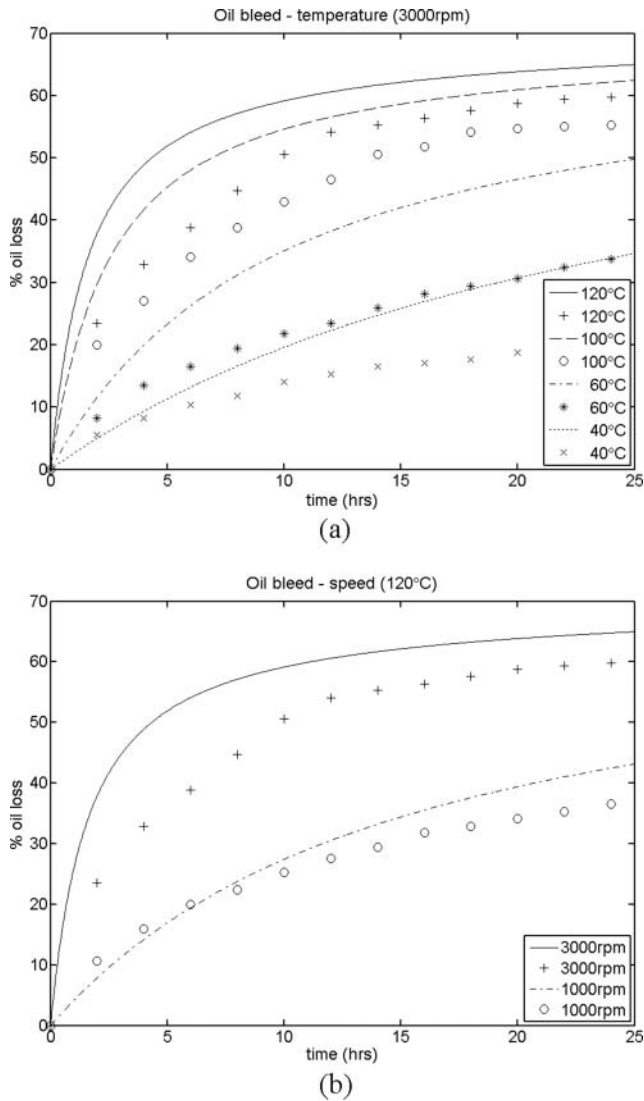


Fig. 9—Oil loss due to oil bleed for an isotropic grease microstructure. The symbols represent the experiments, and the lines represent the model.

Figure 11 shows an extrapolation of the anisotropic model up to 1000 h and includes more temperatures and rotational speeds. All curves finally reach the same maximum percentage of oil loss and give a good indication of the time it takes to reach this maximum. At increased temperatures or rotational speeds, the maximum percentage of oil loss is reached earlier. This could be correlated to relubrication intervals of bearings, which also become shorter when temperatures or speeds are high.

It is anticipated that the oil-bleed model can be used for all greases that have microstructures similar to that of the lithium complex grease studied here. This means that also lithium hydroxystearate greases and calcium greases can be modeled in the same way. However, the grease properties presented in Table 3 will be different and have to be measured for each individual grease.

It must be recognized that the model presented here is not yet complete apart from conditions based on uniform fibers and the homogenous microstructure. Other forces might become rel-

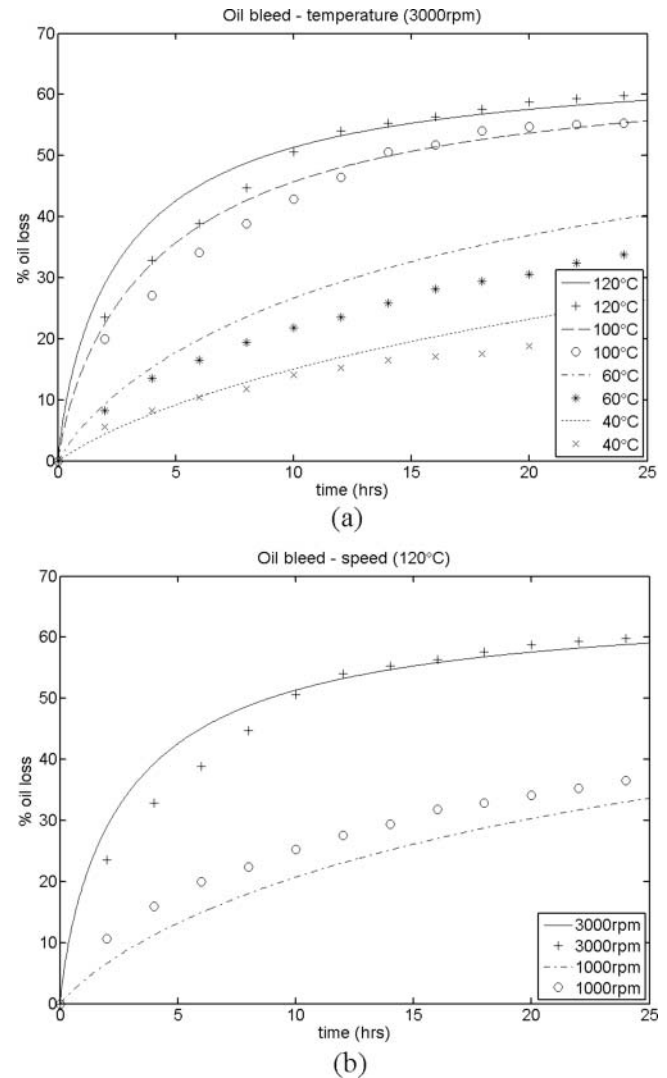


Fig. 10—Oil loss due to oil bleed of an anisotropic grease microstructure where fibers that are initially parallel and then tilt to become perpendicular to the pressure gradient or oil flow. The symbols represent the experiments, and the lines represent the model.

evant at conditions where the centripetal force, which is the driving force in the current model, becomes small. Here one can think of the capillary force acting at the grease–air interface. Including capillary forces into the model results in a reduced oil-bleed rate and in a threshold force below which there is no oil bleed. Indications for capillary effects are found in static grease-bleeding experiments where the temperature was lowered after some time and it was observed that the oil that did bleed out of the grease was sucked back into the grease again. Also, the reduction of the oil-bleed rate found in the current experiment at reduced filling rates of the cup, i.e., grease mass, seems to indicate the existence of capillary forces. It is expected that the dependence on filling rate of the cup will become visible in the model when additional forces like the capillary forces, fiber interactions, and nonhomogeneous microstructures are included. The model also does not include other secondary effects such as the force needed to deform the soap network during bleeding; oxidation of the oil,

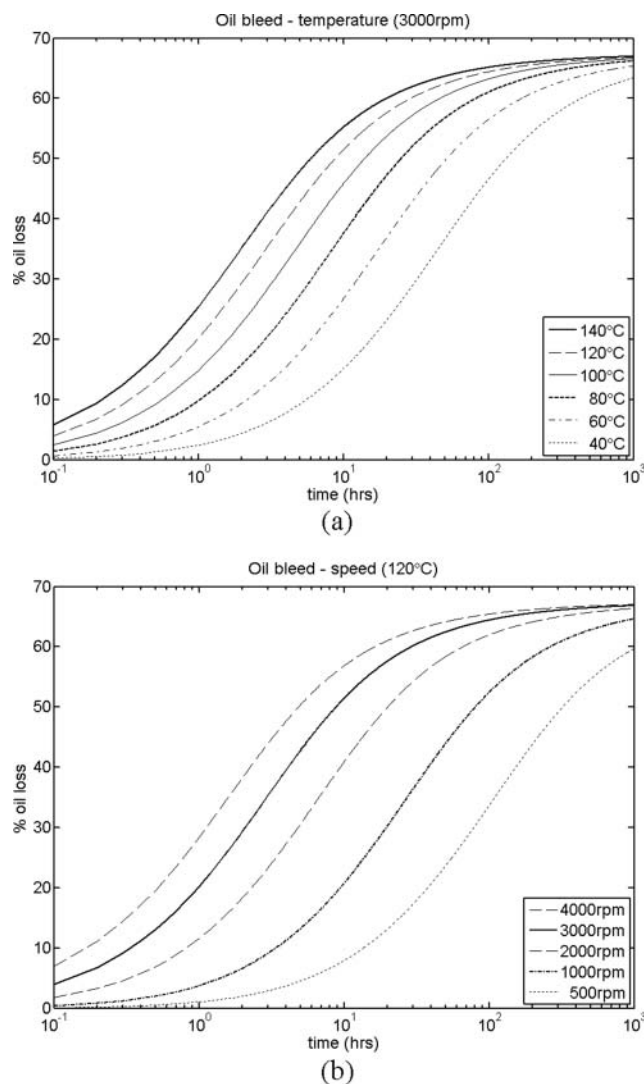


Fig. 11—Oil loss due to oil bleed over a long time span as an indication of relative grease life.

which results in growth of fiber diameter (Salomonsson et al. (6)); and the buoyant force due to the difference in mass density between the thickener fibers and the oil. Including this last effect might lead to an explanation for the oil layer sometimes found on top of the grease after long storage times.

Nevertheless, the current model gives a good indication of the oil-bleed rate when including the anisotropic deformation of the soap microstructure.

CONCLUSION

The microstructure of a lithium complex grease has been visualized using AFM. The images show that the grease consists of solid fibers embedded in oil. These fibers have a twisted rope-like shape with a diameter of $\sim 0.2 \mu\text{m}$.

The oil bleeding of grease has been modeled as viscous flow through a porous soap microstructure subjected to a centripetal body load. The permeability of the soap microstructure has been assumed to be homogeneous and has been derived for an isotropic and anisotropic fiber arrangement. The model results show that the soap microstructure must become anisotropic during oil bleed, i.e., the fibers all tilt perpendicular to the pressure gradient and oil flow, to predict the experimental results and make the oil bleed stop at a certain fiber-volume fraction.

The model has been validated with experiments which have been run over a period of 24 h and can be used with good confidence to simulate the bleeding process over longer periods of time at various speeds and temperatures. The oil-bleed model can be used as input for a replenishment model for grease-lubricated contacts.

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